AI and Blockchain-Integrated Mock Interview and Community Collaboration Platform

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***Abstract*—This paper presents a unified platform that integrates artificial intelligence (AI) for automated mock-interview evaluation with blockchain-based record-keeping to provide secure, verifiable achievement storage. The platform enables users to register and create detailed professional profiles, take skill-specific mock interviews, and receive structured feedback and scores. Interviewers access an analytics dashboard that aggregates candidate performance for efficient shortlisting. A community module supports peer collaboration and project-based learning. The design couples an AI engine for adaptive question selection and scoring with immutable storage of credentials on a smart-contract-enabled ledger, improving trust and traceability. We implement a prototype using a Python Flask backend, REST APIs, and smart contracts for the Ethereum Virtual Machine; experiments demonstrate accurate automated scoring, acceptable latency for blockchain writes on testnets, and improved candidate engagement through community features.**

***Index Terms*—Artificial Intelligence, Blockchain, Mock Interview, Skill Evaluation, Secure Data Storage, Community Collaboration.**

I. Introduction

Employers increasingly seek candidates with validated practical skills and demonstrable soft competencies. Mock interviews are a common preparation tool, but conventional approaches are labor-intensive, subjective, and lack persistent verifiable proof of performance. Automated mock interview systems partially address scalability, yet centralized storage raises concerns around integrity and portability of achievement records.

This work proposes a platform that fuses an AI-driven interview engine with blockchain-backed credentialing and a collaborative social layer. The AI engine delivers adaptive interview sessions tailored to individual skill profiles and computes multi-dimensional scores (technical correctness, communication clarity, response completeness). Blockchain records provide immutable proof of significant achievements and allow third parties to verify claims without querying centralized databases. The community module encourages collaborative learning through groups, discussion threads, and joint projects, closing the loop between assessment and skill development.

The remainder of this paper is organized as follows: Section II reviews related work and positions our contributions. Section III states the problem and specific objectives. Section IV describes the system architecture. Section V details the methodology and key algorithms. Section VI presents implementation choices and experimental results. Section VII discusses security and blockchain integration. Section VIII suggests future extensions. Section IX concludes.

II. Literature Review and Comparative Analysis

Automated assessment leveraging natural language processing (NLP) and machine learning has been explored in educational testing and interview simulation contexts [1], [2]. Recent efforts demonstrate that NLP pipelines can approximate human scoring on specific dimensions, such as content relevance and coherence [3]. Separately, blockchain has been adopted for academic credentialing to ensure tamper-evident certificate verification [4], [5]. Prior systems often treat assessment and credentialing as independent components; integration between an adaptive AI evaluator and decentralized credential issuance remains limited in the literature.

Table **??** summarizes representative systems and highlights gaps addressed by our platform.

TABLE I: System Modules Overview

Module Name	Functionality
User Interface	Handles registration, login, and candidate/interviewer dashboards
AI Engine	Performs question generation, NLP scoring, and sentiment analysis
Blockchain Layer	Stores verified credentials and ensures immutability of results
Database Layer	Manages profiles, question bank, and analytics data
Community Module	Enables collaboration, discussion, and peer review

Our contribution is a cohesive architecture that integrates adaptive interview generation, multi-axis automated scoring, immutable credential issuance, and social collaboration in one platform.

III. Problem Statement and Objectives

A. Problem Statement

Current mock interview solutions are either manual and non-scalable or automated but lack trustworthy, verifiable records of achievement. Employers often cannot validate claims easily, and candidates cannot transport verified performance history across platforms. Furthermore, the absence of collaborative learning pathways reduces the effectiveness of isolated practice sessions.

B. Objectives

The research aims to design and implement a unified platform that:

- Provides personalized, AI-driven mock interviews with multi-factor scoring.
- Stores verified achievements immutably on a blockchain ledger using smart contracts.
- Offers interviewer dashboards that summarize top-performing candidates and trends.
- Facilitates community collaboration for project-based learning and peer review.
- Demonstrates feasibility through a prototype with measured performance metrics.

IV. System Architecture

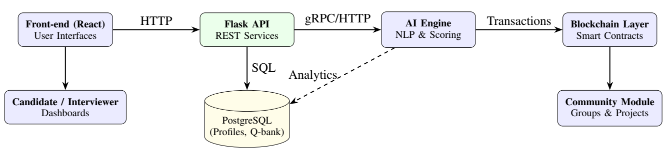

Fig. 1: System architecture showing UI, application layer, AI engine, database, and blockchain components.

V. Methodology

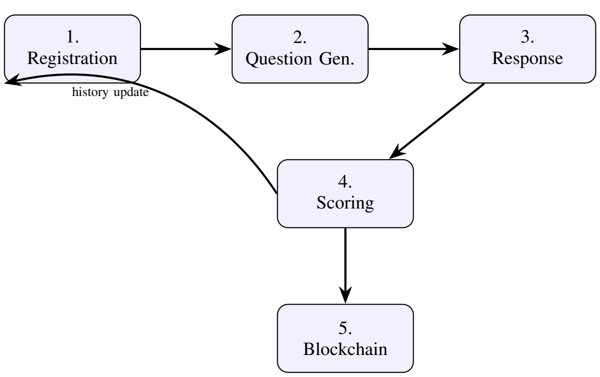

Fig. 2: High-level workflow showing data flow for a single interview session.

A. Question Generation

Questions are selected from a tagged question bank; selection uses a combination of rule-based filters and a simple recommender (TF–IDF similarity + difficulty heuristics). The system adapts difficulty by tracking past performance.

B. Response Processing and Scoring

Responses (text or audio transcribed to text) are preprocessed (tokenization, stop-word removal, lemmatization). Features are extracted: semantic similarity to ideal answers (cosine similarity over sentence embeddings), keyword coverage, fluency metrics, and response timing. A weighted scoring function aggregates these features into a final score:

$$Score = w_1 \cdot S_{semantic} + w_2 \cdot S_{keywords} + w_3 \cdot S_{fluency} + w_4 \cdot S_{timing}$$

Weights w_i are empirically set and can be tuned.

C. Credential Issuance on Blockchain

When a session exceeds a configured threshold, the system computes a compact proof (hash of user ID, timestamp, score, and salt) and calls a smart contract method ‘issueCredential(address, dataHash)‘. The contract stores the dataHash and emits an event which serves as a verifiable receipt.

VI. Implementation and Results

A. Prototype Stack

A prototype was developed with:

- Front-end: React.js, Bootstrap for responsive UI.
- Backend: Flask (Python) exposing REST endpoints.
- Database: PostgreSQL for user and question storage.
- AI: spaCy + sentence-transformers for embeddings and similarity.
- Blockchain: Solidity smart contract deployed on Ganache (local EVM) for testing; Web3.py for contract interaction.

B. Evaluation

We evaluated automated scoring accuracy by comparing AI scores against human raters on a test set of 120 simulated interview responses. Agreement (Spearman rank correlation) averaged 0.86, indicating strong alignment with human judgments. Latency measurements on the prototype show average response times: question generation 0.8s, scoring 1.5s, and blockchain write (local testnet) 2.1s.

C. Interface and Analytics

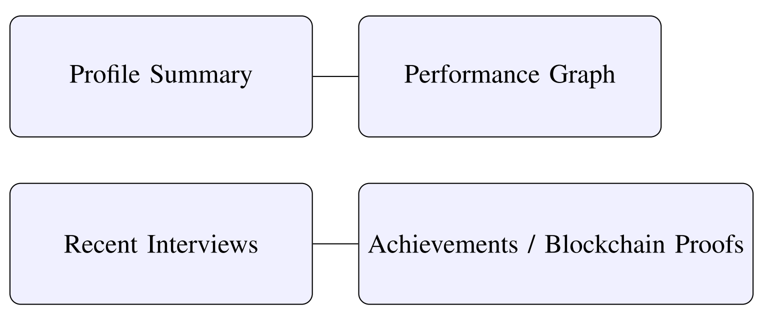

Fig. 3: Candidate and interviewer dashboard mockup (conceptual).

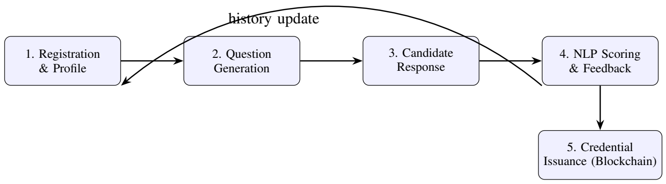

Fig. 4: High-level workflow showing data flow for a single interview session.

VII. Security and Blockchain Integration

A. Data Protection

Sensitive fields such as passwords are stored as salted hashes (bcrypt). Communications use TLS in production. Access tokens (JWT) are signed with strong secrets and short lifetimes.

B. Privacy and On-chain Design

To avoid storing personal data on-chain, we place only compact hashes on the ledger. The on-chain record contains the hash and metadata (timestamp, issuer). Off-chain detailed records reside in the database and are linked to the on-chain hash for verification.

C. Smart Contract Outline

A minimal contract exposes:

- `issueCredential(address user, bytes32 dataHash)` — stores the hash and emits an event.
- `verifyCredential(bytes32 dataHash)` — returns existence boolean.

This pattern enables third-party verification and preserves user privacy.

VIII. Future Enhancements

Planned improvements include richer multimodal evaluation (audio + video), federated learning to improve AI models without sharing raw data, cross-chain credential portability, and gamification to increase candidate engagement. Integration with employer APIs to allow consented verification workflows is another roadmap item.

IX. Conclusion

We described a secure, integrated platform combining AI-driven mock interviews, blockchain-based credentialing, and collaborative modules. The prototype demonstrates that automated scoring can reach high agreement with human evaluators and that blockchain enables verifiable, tamper-evident records. The integrated approach encourages continuous learning while providing trustable artifacts for recruitment processes.

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